CRP 4680/5680: Introduction to Urban Data Science

Introduction to spatial statistics – Spatial autocorrelation

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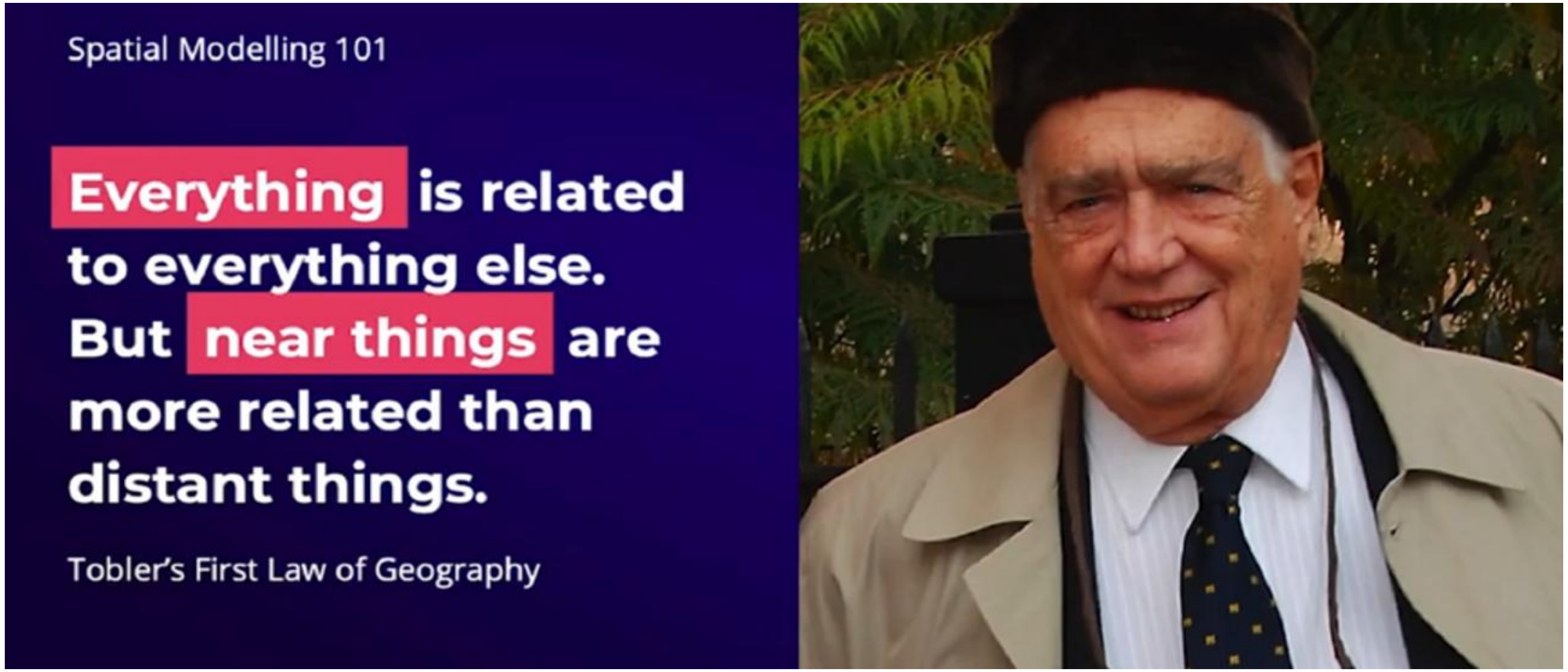
Spatial dependence/Spatial autocorrelation

- Geographic data tend to be spatially dependent.
- ***Spatial dependence*** is "the propensity for nearby locations to influence each other and to possess similar attributes" (Goodchild, 1992, p.33).
- **Spatial autocorrelation** is the **statistical measure** of spatial dependence



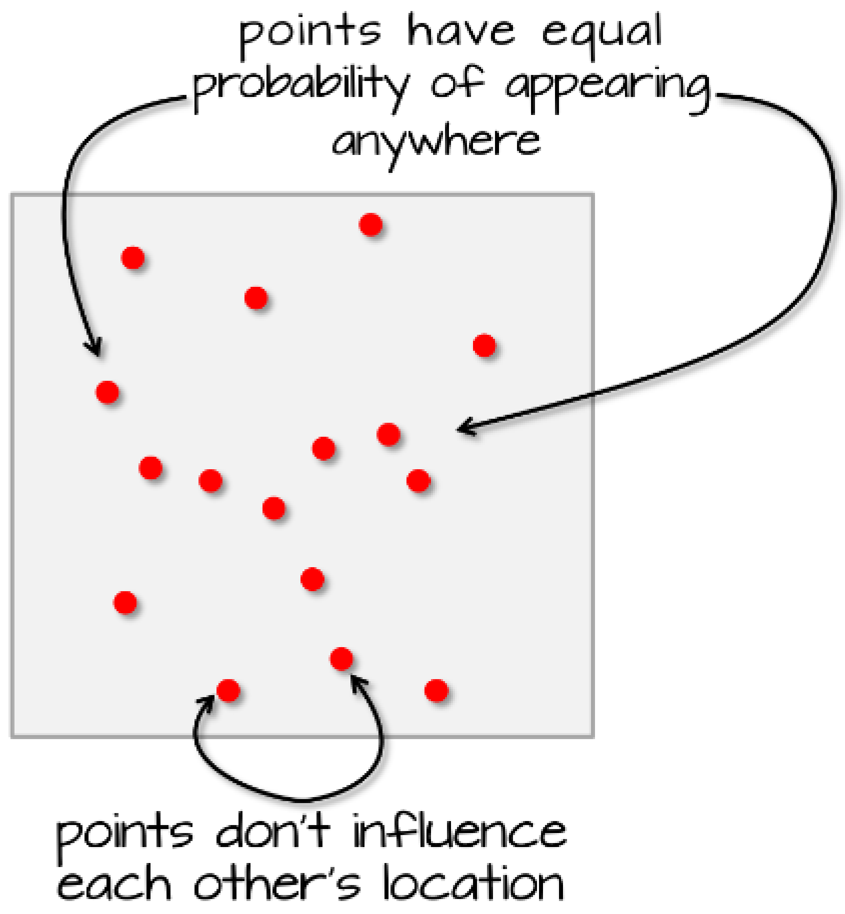
Complete spatial randomness (CSR)

Spatial dependence is a *relative* concept. To quantify the degree of spatial dependence via spatial autocorrelation measures, we compare observed spatial patterns to a *completely random process (CSR)* as a baseline.

Two main principles for CSR:

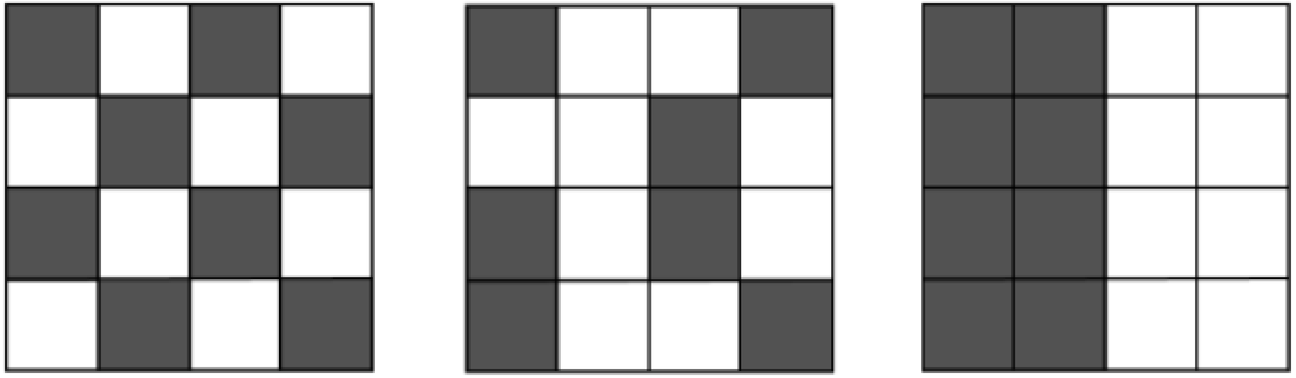
1. Equal probability: any feature has equal probability of being in any position

2. Independence: the positioning of any feature is independent of the positioning of any other feature.



Spatial Autocorrelation – measures of spatial dependence

- Spatial autocorrelation occurs when nearby values are more similar (or dissimilar) than expected under a completely random spatial process **->** spatial **independence**
  - **Positive** spatial autocorrelation → Similar values cluster together (e.g., rich neighborhoods near other rich areas). Most social phenomena are positively spatially autocorrelated
  - **negative** spatial autocorrelation → When events are dispersed (or Dissimilar values are clustered together)



Negative spatial autocorrelation — No spatial autocorrelation — Positive spatial autocorrelation

How can we measure spatial autocorrelation? Let's explore metrics: Moran's I

Global indicator of Spatial Autocorrelation

Moran's I statistic: a spatial statistic measures ***overall spatial autocorrelation*** of a dataset--whether similar values are clustered, dispersed, or randomly distributed

- measures spatial autocorrelation based on both feature locations and feature values.
- Calculation based on the variable X and the "***spatial lag***" of X formed by averaging all the values of X for the neighbors.
- Neighbors' information are stored in a spatial weight matrix.

Moran's *I* is defined as

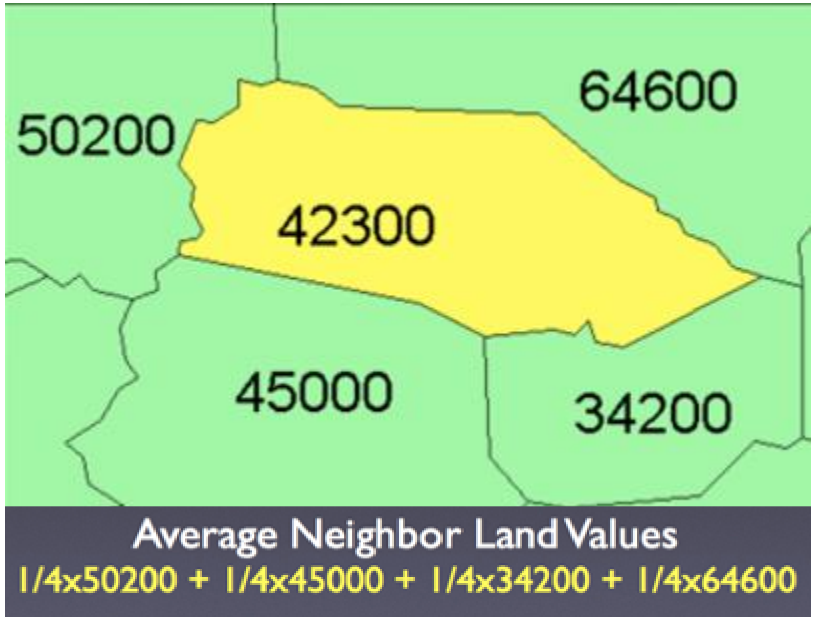
$$I = \frac{N}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij}(X_i - \bar{X})(X_j - \bar{X})}{\sum_i (X_i - \bar{X})^2}$$

where N is the number of spatial units indexed by i and j; X is the variable of interest; $\bar{X}$ is the mean of X; and w_{ij} is an element of a matrix of spatial weights.

Spatial lag

- Spatial lag is like a local weighted average of the values surrounding *i*.
- A way to link your focal location i to its neighbors.

$$WX = \sum_j w_{ij} X_j$$



Why local <u>weighted average?</u>

What is the problem?

- Location A: 3 neighbors -> a total weight sum of 3
- Location B: 8 neighbors -> a total weight sum of 8.
- The unit with more neighbors exerts more influence. So, A has more influence than B.

1 A	5	3	7
9	2 B	4	6
10	4	3	5
8	2	1	8

Solution?

- ***Row Standardization***: each neighbor's weight is divided by the total number of neighbors. Both units (A or B), regardless of neighbor count, have a total weight sum of 1.
- Sum of all weights in row i equal to 1

row-standardized weight:

$$w'_{ij} = \frac{w_{ij}}{\sum_j w_{ij}}$$

Spatial lag

- Spatial lag is like a local weighted average of the values surrounding *i.*

$$WX = \sum_j w_{ij} X_j$$

Rook row-standardized weights:

- For A, weights are: 1/3,1/3,1/3
- For B, weights are: 1/8,1/8,1/8, 1/8,1/8,1/8, 1/8,1/8

- Using Rook row-standardized weights, what is the spatial lag of A? Or B?

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Spatial lag

- Spatial lag is like a local weighted average of the values surrounding *i.*

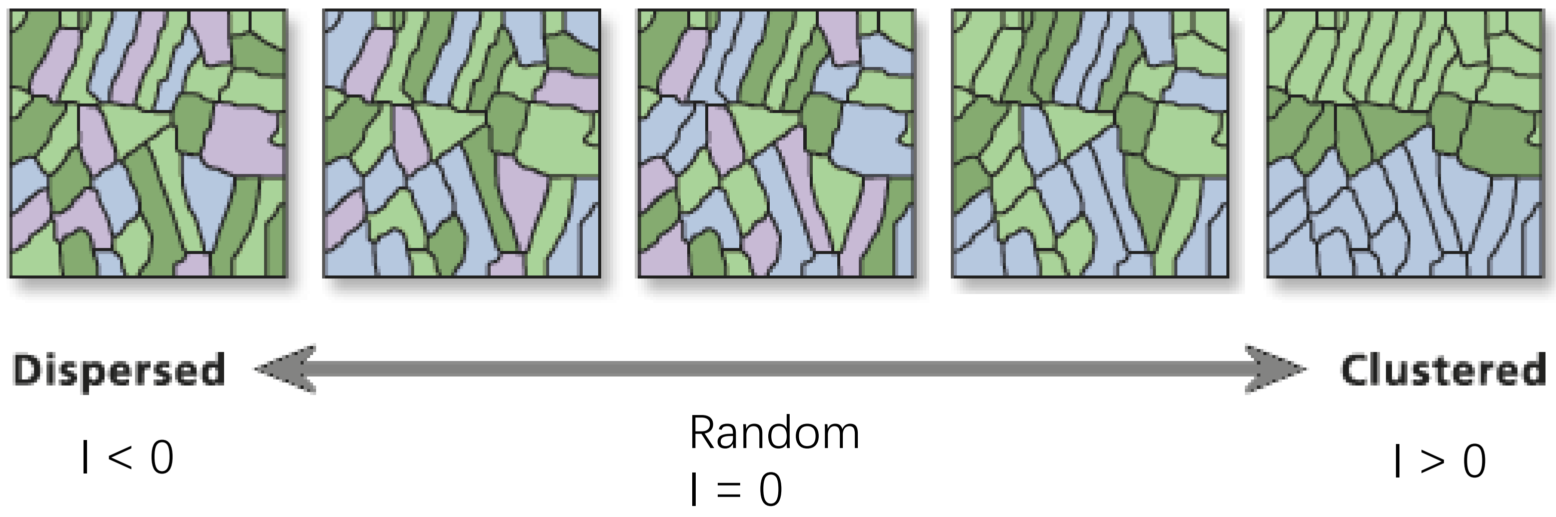
$$WX = \sum_j w_{ij} X_j$$

- In other words, using row-standardization, you try to use the average housing price of your neighbors as reference, while without the row-standardization, you use the sum of housing prices of all neighbors as reference!

- Spatial lag forms the foundations of how we understand spatial autocorrelation.
- It allows us to understand this question: Are my neighbors similar to me or not?

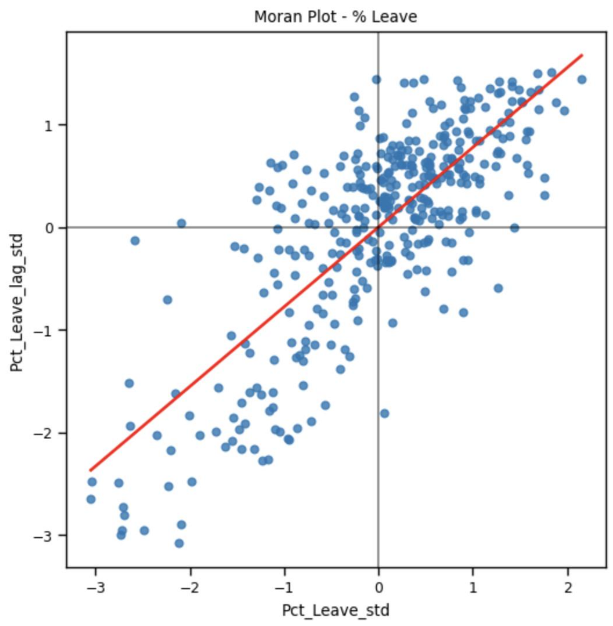
Interpreting Moran’s I

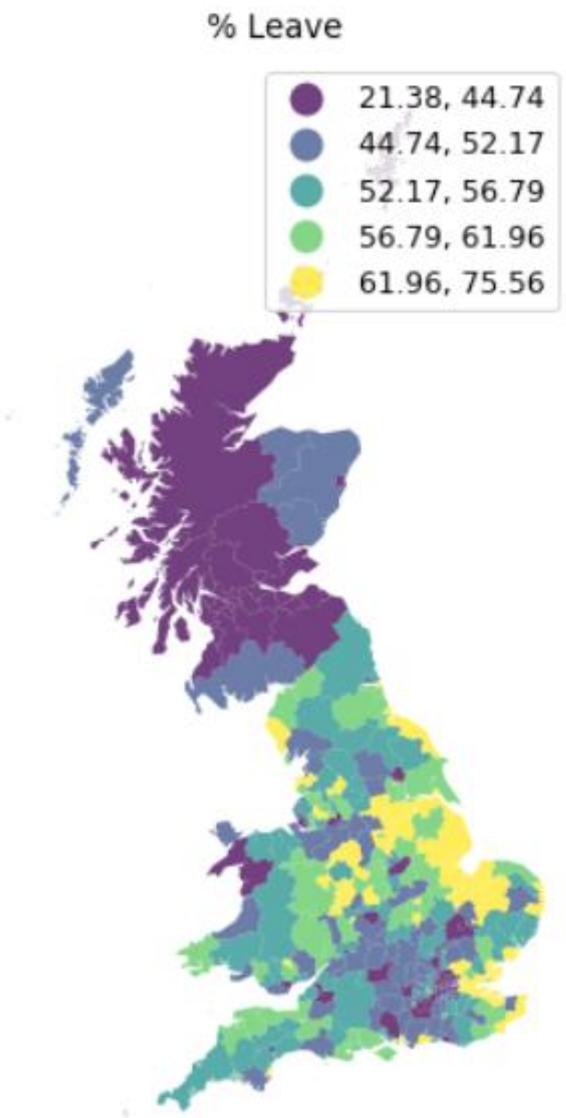
- Moran’s I evaluates whether the pattern is clustered, dispersed, or random
- a positive Moran's I index value indicates tendency toward clustering while a negative Moran's I index value indicates tendency toward dispersion. (ranges from -1 to 1).
- Moran's I = 0 (the attribute being analyzed is randomly distributed among the features in your study area (*Complete Spatial randomness*)



Global Moran's Plot

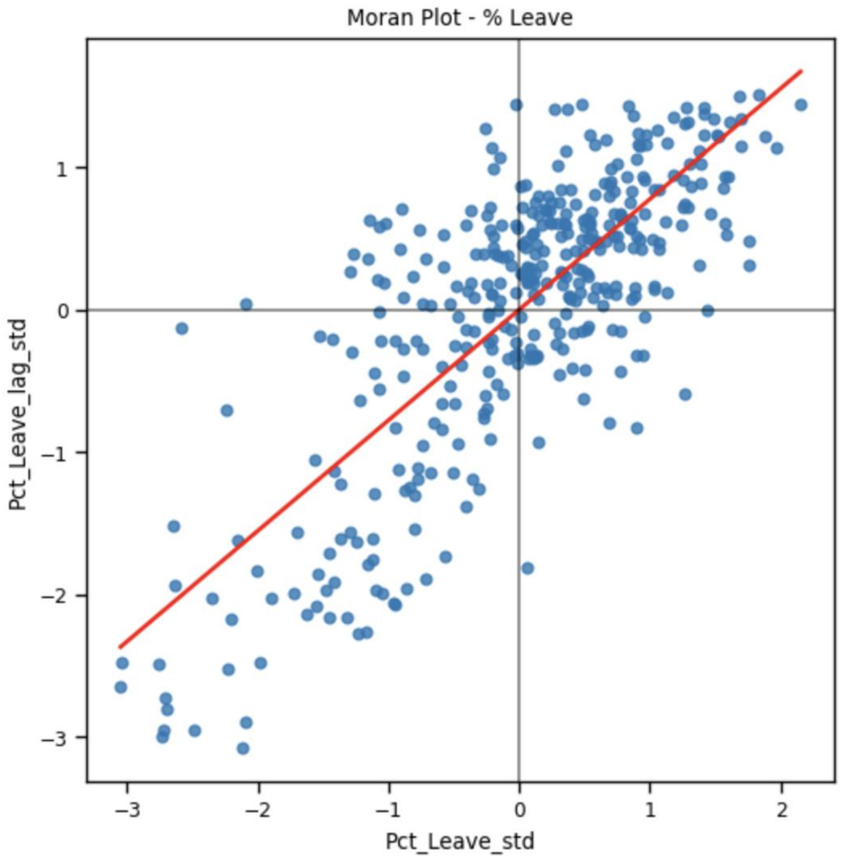
- This figure shows the locations and percentages that voted to leave the EU.
- This chart is a Moran's Plot and shows each observation's vote compared to its spatial lag vote (standardized).





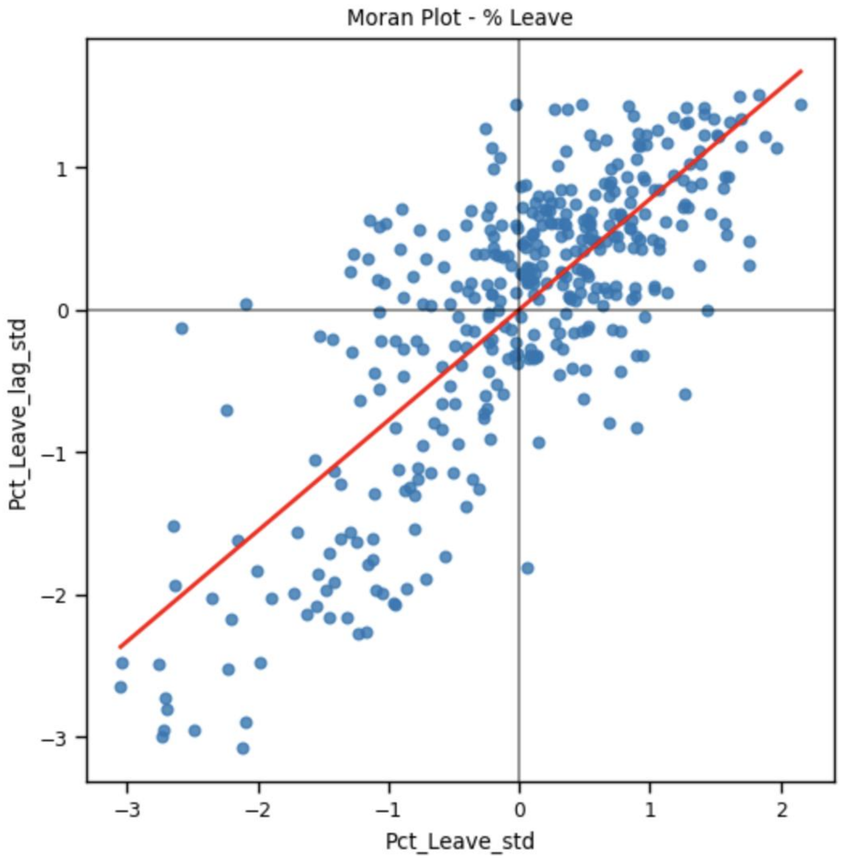
Global Moran’s Plot

- This chart is a Moran’s Plot and shows each observation’s vote compared to its spatial lag vote (standardized).
- Is there positive of negative spatial autocorrelation?



Global Moran's Plot

- This chart is a Moran's Plot and shows each observation's vote compared to its spatial lag vote (standardized).
- Is there positive of negative spatial autocorrelation?
- The leave votes are spatial autocorrelated in a positive way.



Moran's I (Global)

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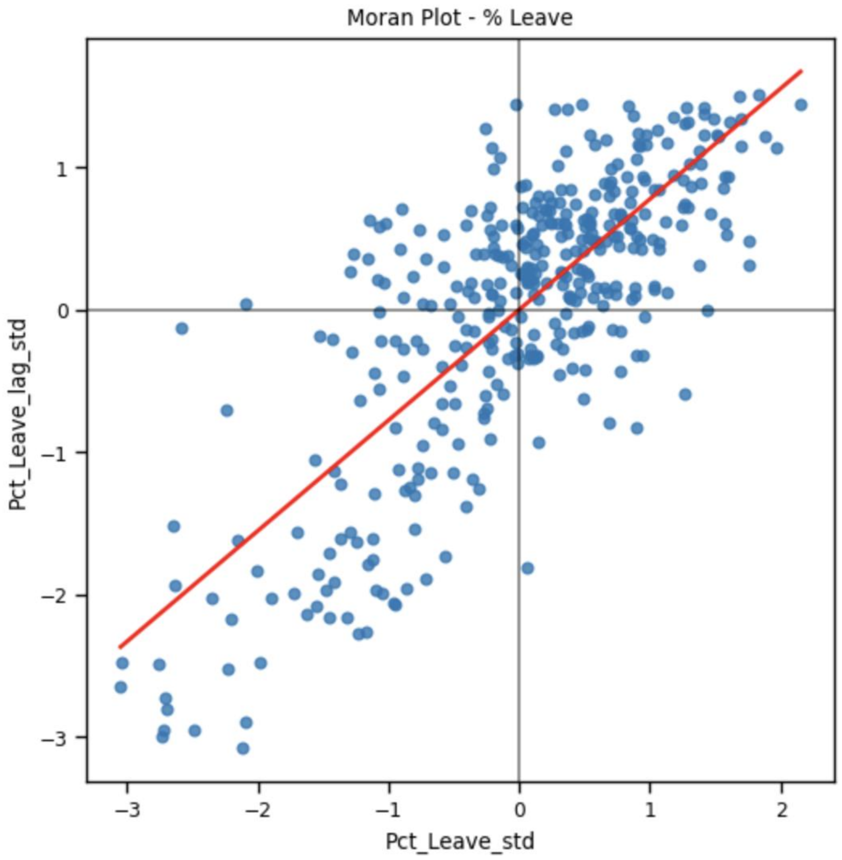
- Is there positive of negative spatial autocorrelation?
- The leave votes are spatial autocorrelated in a positive way.
- **The Moran's I corresponds to the slope of this red line.**



Moran’s I (Global)

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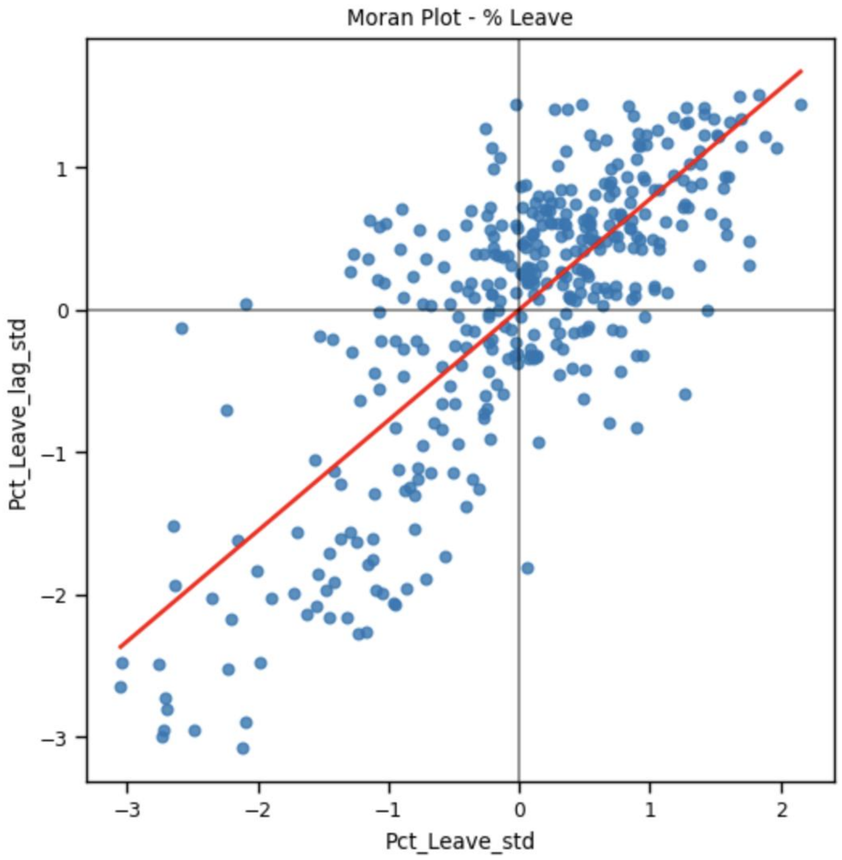
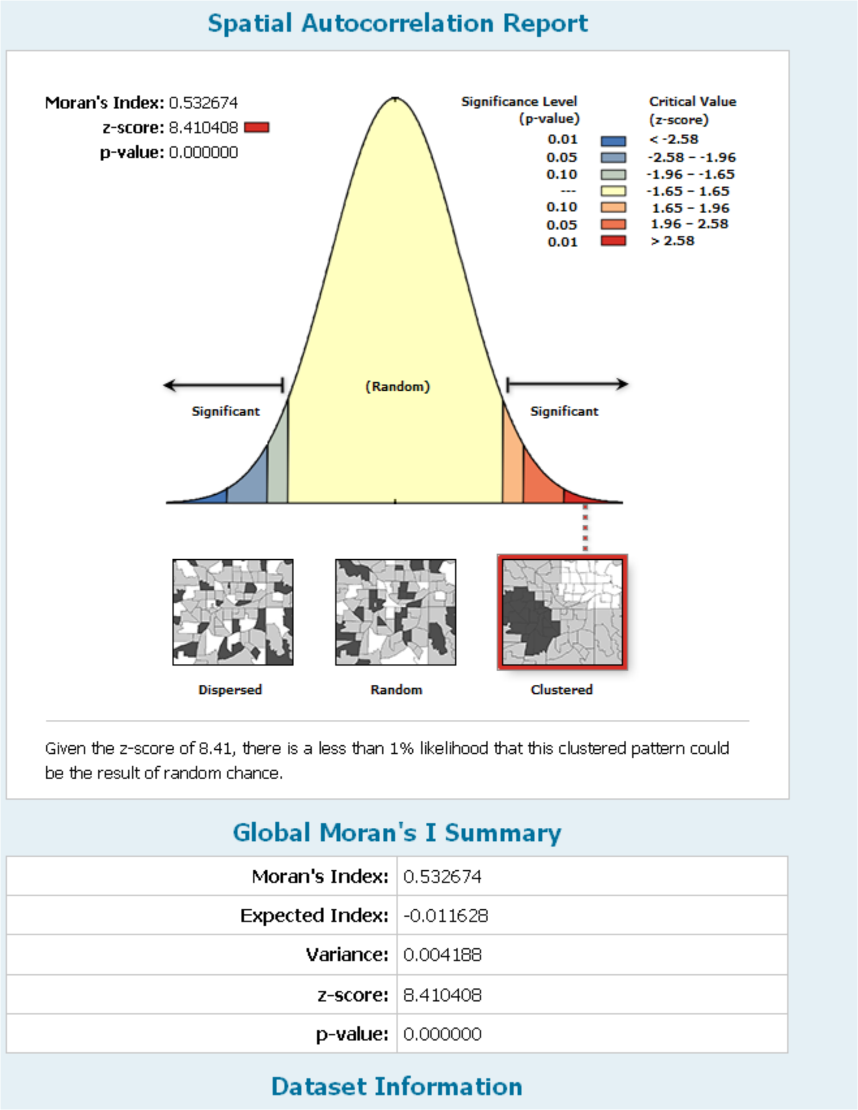
- Is there positive of negative spatial autocorrelation?
- The leave votes are spatial autocorrelated in a positive way.
- **The Moran’s I corresponds to the slop of this red line.**
- What would this picture look like if the Moran’s I were negative?



Interpreting Global Moran’s I



Moran’s Index: 0.532674 — positive Moran's I value indicates a strong clustering pattern, meaning similar values (e.g., high or low) tend to be near each other.

Significance Level: This report displays the distribution curve, with significant results shown at the tail ends. The clustering pattern falls in the right tail.

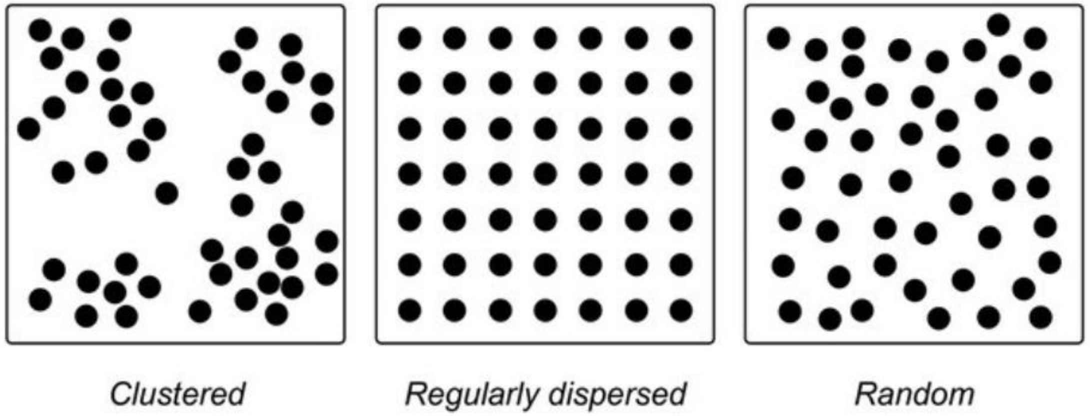
- Z-Score: 8.410408 — A high z-score (well above 1.96) implies that the clustering is far from random (at least at the 5% significance level).
- P-Value: 0.000000 — A very low p-value (< 0.01) indicates that there is less than a 1% likelihood that this observed clustering pattern is due to random chance (statistical significance).

Expected Index and Variance: The expected Moran's I index (-0.011628) and variance (0.004188) serve as baselines to compare the observed Moran's I, The observed value is considerably higher than expected under random spatial distribution.

Inferential Spatial Statistics
Null and Alternative Hypotheses

- Null Hypothesis:
  - The spatial pattern is <u>**random**</u>
  - CSR (complete spatial randomness): events or values are distributed **independently and uniformly** across a given area, with no underlying spatial structure, clustering, or dispersion beyond what would be expected by pure chance.

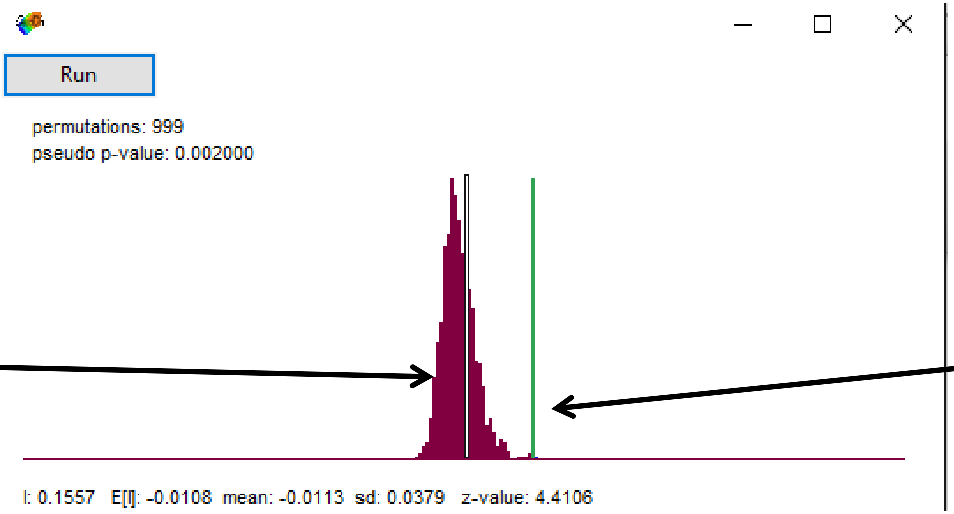
- Alternative Hypothesis:
  - The spatial pattern is <u>**not**</u> **random**
  - It may be clustered or dispersed



Spatial Statistical Hypothesis Testing(Global Moran’s I)

- Because of the complexity of spatial processes, it is often difficult to derive a test statistic with a known probability distribution
- Instead, we often use a Monte Carlo simulation: We take multiple samples from a **random** spatial pattern. Moran’s I is calculated for each sample, and then a frequency distribution is drawn
- This *simulated sampling distribution* is used to measure the probability of obtaining our actual observed spatial statistic

Empirical frequency distribution from 1000 random patterns (“samples”)

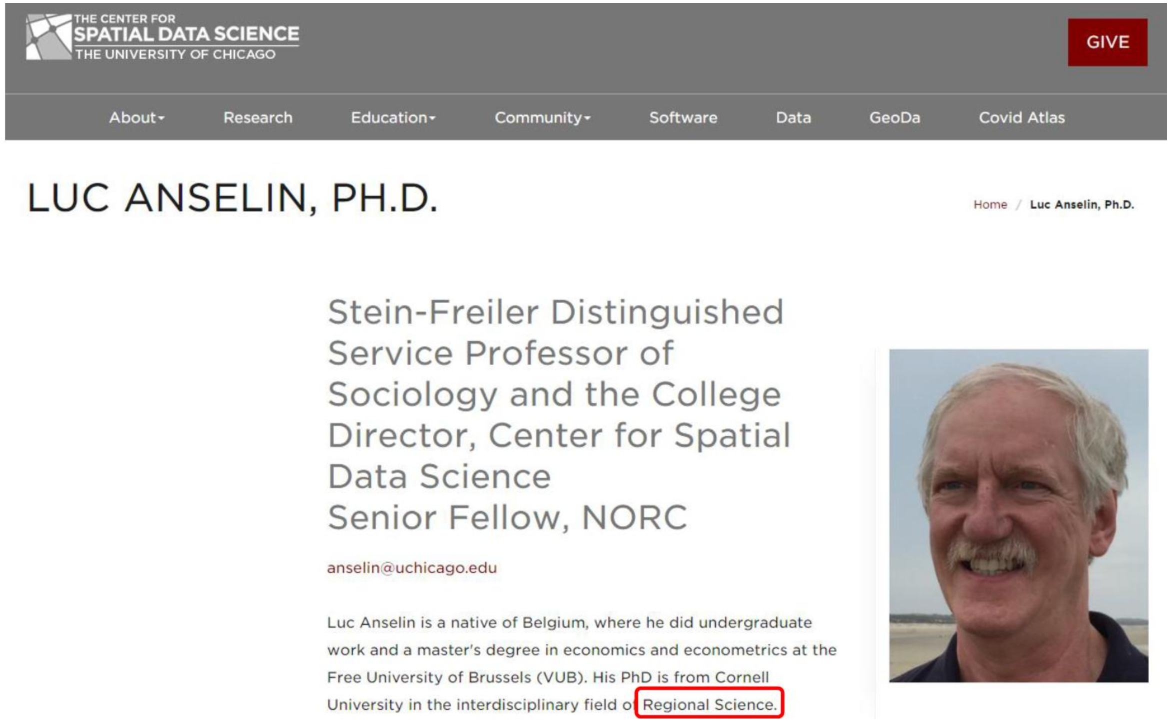


Our observed value:
--highly unlikely to have occurred if the process was random
--conclude that process is not random

Local measures of clustering?

- We have a global sense of clustering from Moran's I, but lack information about its local variation
  - Which areas contribute the most to our global statistics? Where are the clusters? Which areas not at all?

- **Local Indicator of spatial autocorrelation (LISA)**
  - The Local Moran statistic (Anselin, 1995): as a way to identify local clusters and local spatial outliers.
  - Calculates Local Moran's I for each spatial unit and evaluating the statistical significance for each based on its similarity to its neighbor



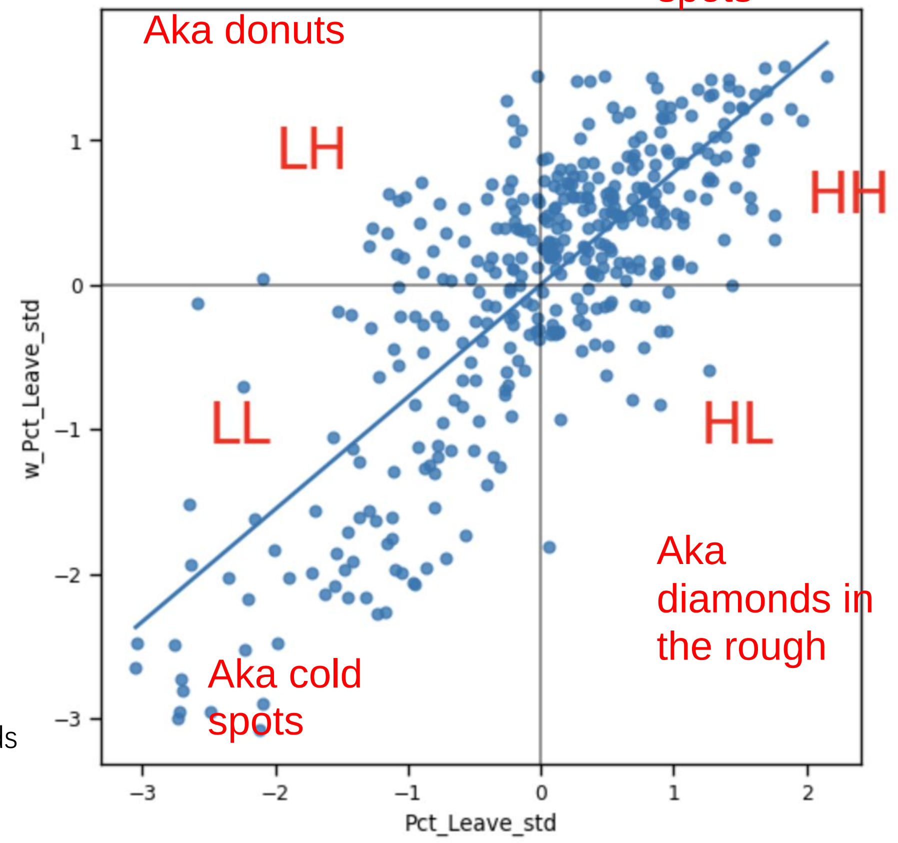
Moran’s I (Local)

- Local indicators of spatial autocorrelation (LISA) are NOT summary statistics but tell us about the structure of the data.
- Local Moran’s I tells us whether our observation is HH, LL (both are positive) or LH, HL (both are negative)

The central park in the NYC

Aka hot spots

Wealthy neighborhoods surrounded by other wealthy neighborhoods



Poor neighborhoods clustered together

A high-priced property in a low-income neighborhood

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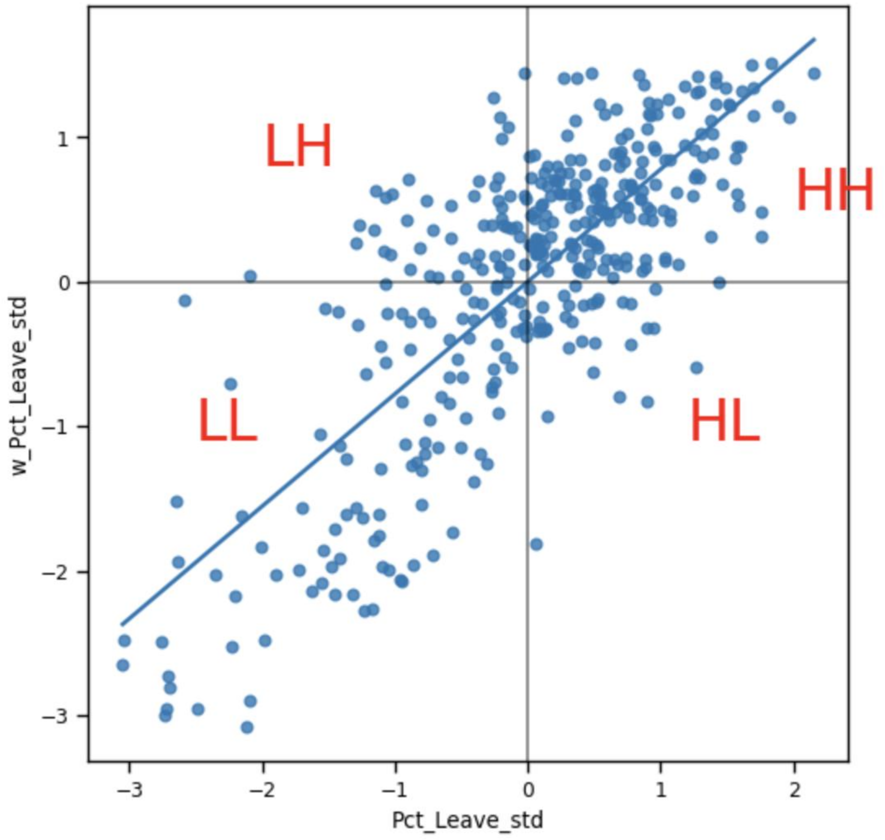
Observation Neighbor

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j \; ; \; m_2 = \frac{\sum_i z_i^2}{n}$$

Variance of the data



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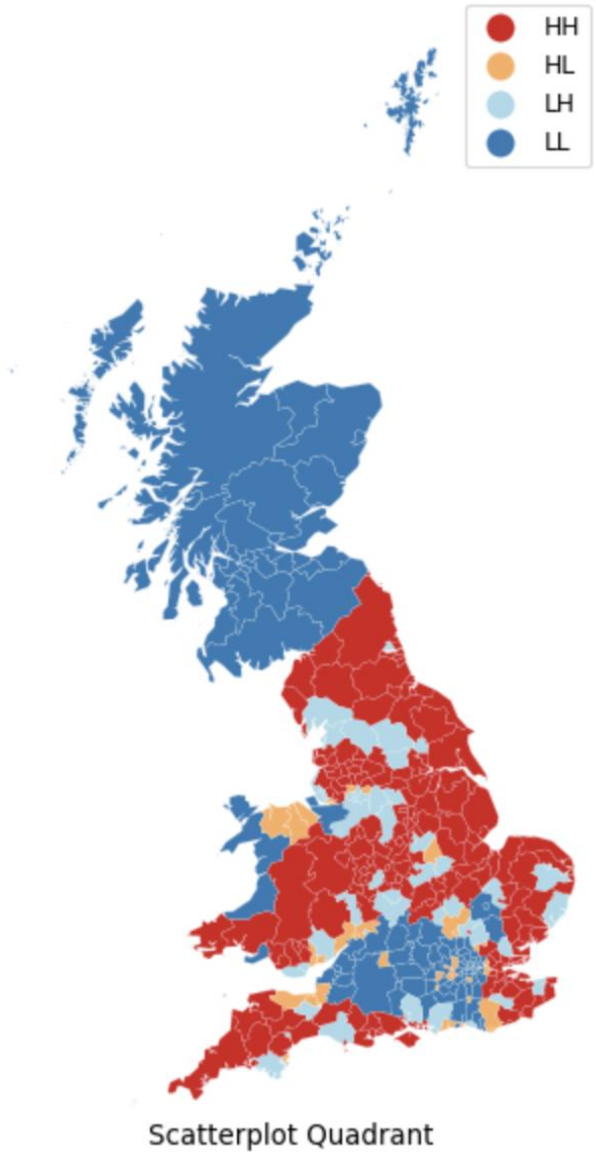
- Map of the labels



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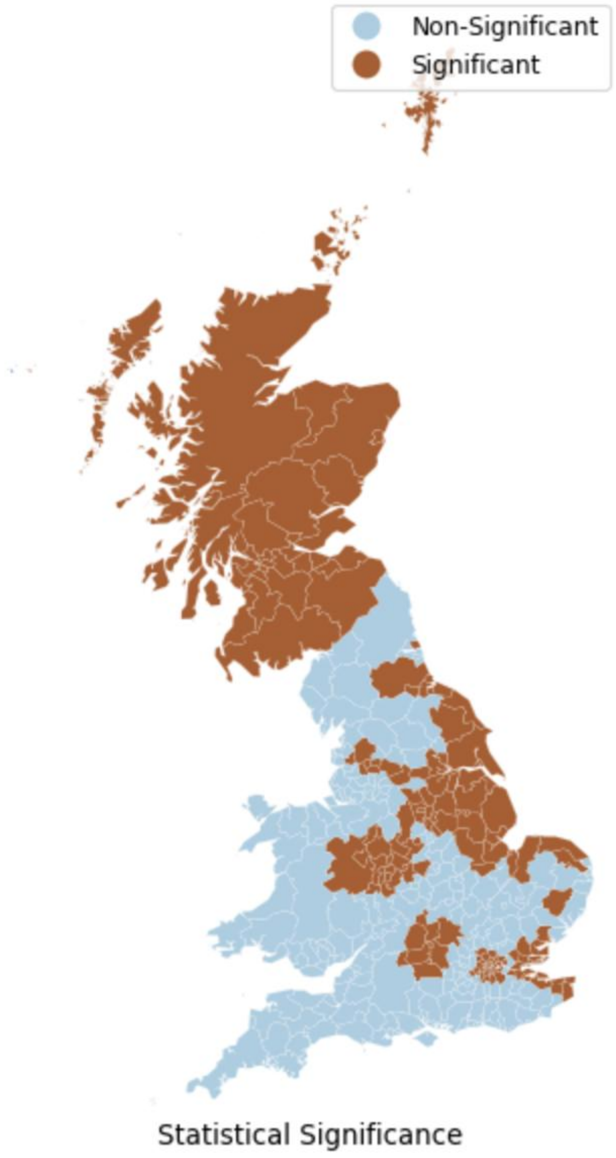
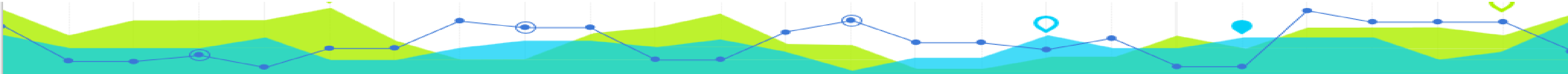
- We need to test for the significance!





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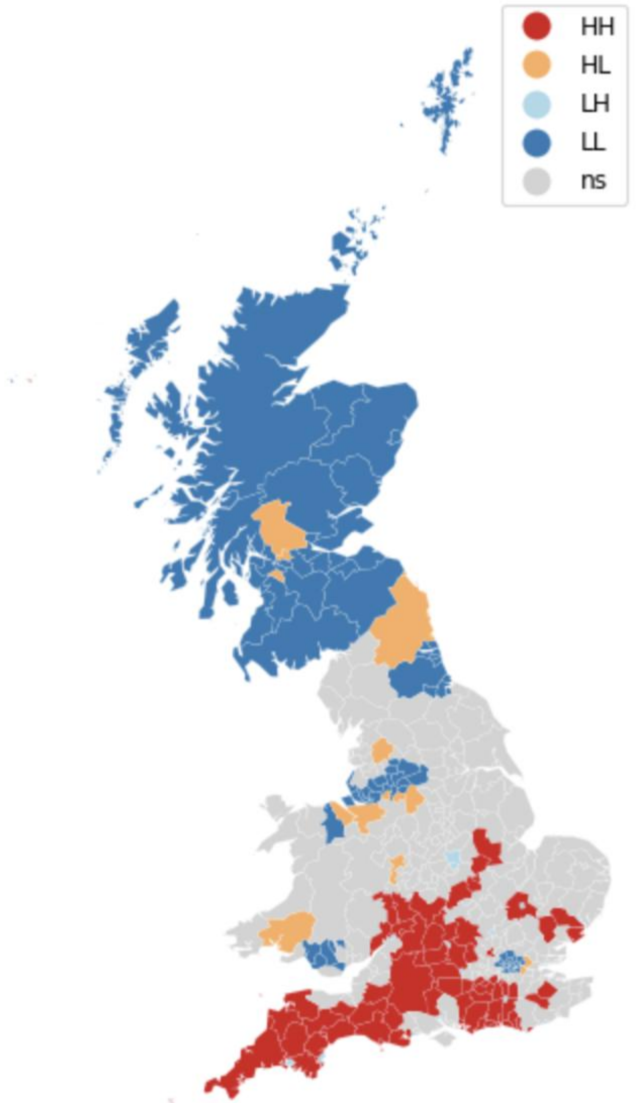
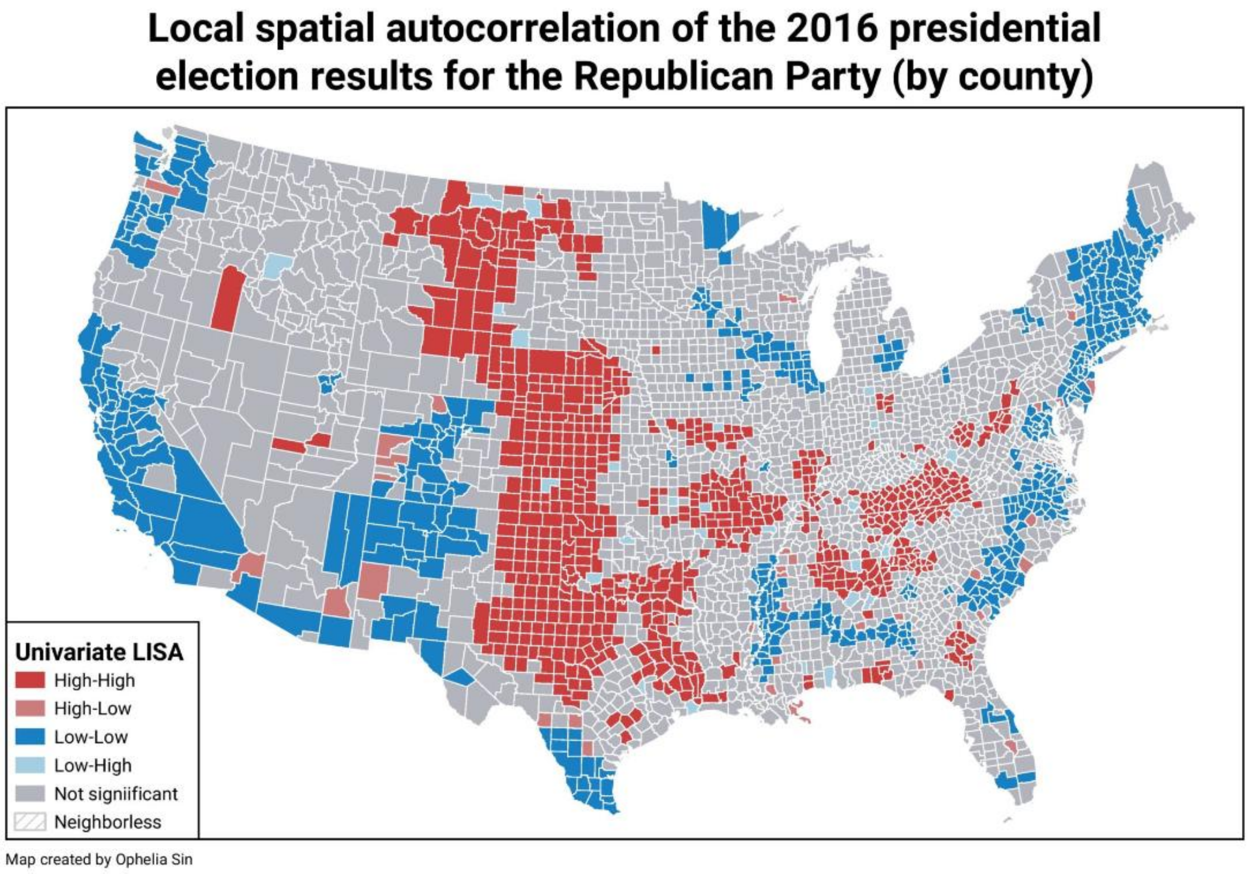
- Map of the labels
- We need to test for the significance!
- This is what we get when we filter for only significant observations



Moran Cluster Map

Local Indicator of spatial autocorrelation interpretation



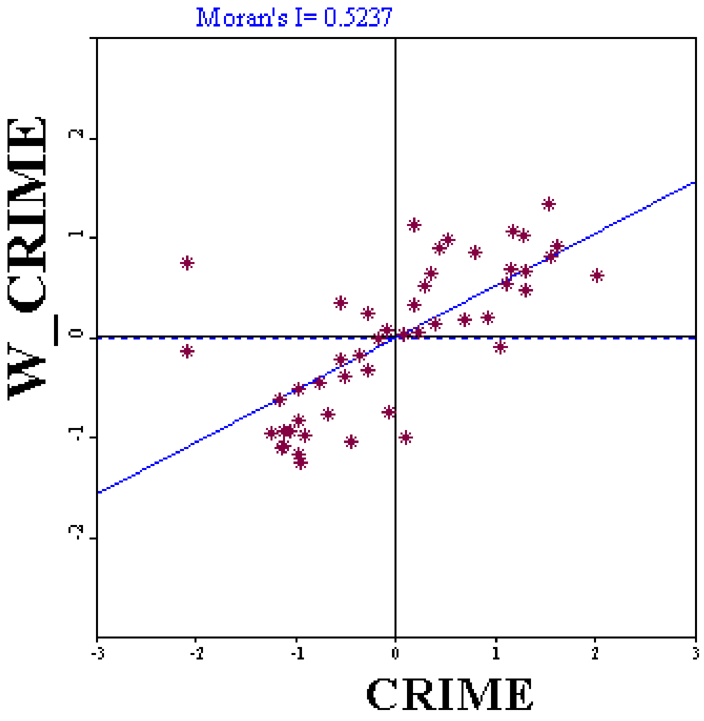
- High-High (red): Counties in red are Republican strongholds where high Republican voting is surrounded by other high Republican-voting counties, indicating clusters of strong Republican support.
- Low-Low (blue): Counties in blue show clusters of low Republican voting surrounded by other low-Republican counties, representing Democratic-leaning regions,
- High-Low (light red): These counties have high Republican voting but are surrounded by counties with lower Republican voting, indicating isolated Republican support in less supportive regions.
- Low-High (light blue): These counties show low Republican voting but are surrounded by higher Republican-voting counties, typically isolated Democratic-leaning areas within Republican regions.
- Not Significant (gray): Gray counties show no statistically significant clustering pattern, suggesting a mix or lack of strong spatial autocorrelation.

LISA Scatter Plots

- Moran's I can be interpreted as the correlation between variable X and the "spatial lag" of X formed by averaging all the values of X for the neighboring polygons
- We can then draw a scatter diagram between these two variables: **X** and **lag-X**
- The slope of the regression line is Moran's I

Spatial outliers: **Low/High**
negative SA

Spatial Clusters: **High/High**
positive SA



Each quadrant corresponds to one of the four different types of spatial association

Spatial Clusters: **Low/Low**
positive SA

Spatial outliers: **High/Low**
negative SA